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Diploma in Engineering (Sixth Semester) Examination, Nov.-Dec., 2015

(Mech. Engg. Branch)

POWER PLANT ENGINEERING

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are compulsory unless mentioned otherwise. Use of steam table and Mollier chart is allowed. Assume suitable data whenever necessary. In case of any doubt or dispute the English version question should be treated as final*

1. A power plant using steam as working fluid operates on a Rankine cycle. The boiler and condenser pressure are 30 bar and 1 bar. The condition of steam entering the

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prime mover is dry and saturated. Find the thermal efficiency of the cycle.

(i) Neglecting the feed pump work.

(ii) Considering the feed pump work. 10

Or

Explain with the help of H-S and T-S diagram various methods of improving efficiency of Rankine cycle.

2. Explain Lamonth Boiler with neat sketch. 10

Or

Explain Super heater and Economiser with neat sketch.

3. The inlet conditions to a steam nozzle are 10 bar and 250°C exit pressure is 2 bar assuming isentropic expansion and negligible inlet velocity determine :

(i) throat area

(ii) exit velocity.

(iii) exit area of nozzle for maximum discharge. 10

4. Explain jet and surface condenser with diagram. 10

Or

Explain Cooling towers with diagram.

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5. Explain controlling factor in load variation. 10

Or

List out the material used for the pipes in steam power stations.

6. Explain different moderator in nuclear power plant. 10

Or

Describe the working of a fast Breeder reactor.

7. Describe component and fuel injection system for Diesel engine power plant. 10

Or

Explain cooling and lubrication system for Diesel engine power plant.

8. Explain open and closed cycle used in gas turbine plant by P-Y and T-S diagram. 10

Or

Explain method of improving the thermal efficiency of gas turbine power plant.

9. Draw a neat diagram of Hydro-electric power plant. Explain the function of its elements. 10

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10. Define : 10

(i) Connected load,

(ii) Maximum demand,

(iii) Average load,

(iv) Load factor,

(v) Demand factor,

(vi) Diversity factor,

(vii) Plant capacity factor,

(viii) Plant use factor.

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